Visualising and Forecasting Stocks using XGBoost Algorithm

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Abstract — Investing in the stock market has always been a popular way for people to generate high returns. Nowadays, even a common man can invest in the stock market and generate good profits if he/she analyses the previous data correctly. Hence, extensive research is needed in the methodology of stock price movements to predict stock prices accurately. The literature survey of this study explored other algorithms like linear regression, pace regression, sentiment analysis, etc for the same. This paper explores using the XGBoost algorithm for the price prediction of stocks in the Indian stock market due to its better accuracy than other models. The system integrates the XGBoost algorithm to train the model on Indian stock market data. The uniqueness of this system is that it entails the use of the XGBoost algorithm, a seldom-used algorithm for predicting stock prices in the Indian stock market. The proposed system also has the distinct characteristic inclusion of the Indian stock market data which is rarely seen in many stock price prediction systems. The accuracy of the model obtained is approximately 84.42% on average. By referencing the research literature, this study also explores other machine-learning techniques. It concludes that though the XGBoost model is highly accurate in many instances, it is best used in conjunction with different models to enhance their utility.

Keywords—XGBoost, Machine Learning algorithms, Stock forecasting, Stock Price Prediction, Yfinance API, Python Dash framework

I. Introduction

The main idea of the stock market, since its inception, has been to generate profits for the investors. Hence, it has been a mission for both researchers and investors to predict the stock price of various stocks available in the stock market accurately. This demand for a high level of accuracy has given rise to various machine learning and optimization algorithms.[12] Although common people and businessmen have been investing in company securities since the 18th and 19th centuries, the stock market trend became popular worldwide after the First and the Second World Wars. However in India, the trend of 'stock marketing' was popularised in the 1990s after the liberalisation of our economy. This initiated in the stock market, a demand for tools that could accurately predict the trends of stocks and their prices. This demand thus gave rise to increased research in machine learning and deep learning algorithms to forecast stock prices.[10,12] Two analysis methods exist for analysing stock prices: Fundamental Analysis and Technical Analysis. Fundamental analysis deals with a particular stock index's economic, industrial, and company analysis. This technique includes the study of the latest geopolitical and financial news, the financial deals made by a particular company or general rates of various goods and services in the market. Technical analysis is a method in which the future price of a stock is predicted using its previous performance data only. Predictions are made by exploiting implications hidden in past trading activities and by analysing patterns and trends shown in price and volume charts.[1,10,20] Sureshkumar et al. also explore various machine learning algorithms such as linear regression, least mean square, isotonic regression, simple linear regression, pace regression, SMO regression, multilayer regression, and Gaussian processes with varying levels of accuracy while Nayak et al. and Wuthrich et al. explore stock price prediction using sentiment analysis of news semantics.[1,8,10]This research paper advocates for the XGBoost algorithm as being one of the most successful algorithms for predicting future stock prices. Thus, the proposed system uses the method of technical analysis to determine the future price of a particular stock. This research uses NSE's Nifty 50 companies' stock prices to train and test the ML(Machine Learning) model. The primary goal is to demonstrate how integrating historical data with predictive models can offer practical, data-driven insights for investors, aiding in more informed decision-making. The literature referred to in this paper was majorly focused on the use of linear regression and random forest algorithm. The uniqueness of the proposed system is that it integrates two of the most seldom-found components into it, i.e., the XGBoost algorithm and the Indian stock market data.[9, 15]

A. Stock Forecasting:

It is also known as stock market prediction, it is the process of predicting the future value of the stock.

B. Stock Price Prediction:

It predicts the future value of an individual stock.

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C. Yfinance:

It is an open-source library that helps to access the financial data available on Yahoo Finance.

D. Python Dash:

It is an open-source tool used for building web applications using Python.

II. Problem Statement

Investors struggle with the absence of tools that can process large datasets of historical stock data, market forecasts, and dynamic changes in the market. Standard approaches seldom take advantage of the cutting-edge algorithms available for grand-scale data. The accuracies of these algorithms we found in the literature survey, like pace regression, sentiment analysis and linear regression were too low for real-time use in stock price prediction. Thus, there is a strong need for highly efficient ML models to predict stock prices with good accuracy. Furthermore, current models for predicting stocks generally do not consider the Indian stock market, leading to a lack of resources for accurate stock forecasting and stock price prediction. Also, the systems in place for predicting stock market prices for the Indian stock market do not use the consistently efficient XGBoost algorithm. This paper, hence, focuses on using the XGBoost algorithm on the Indian stock market to predict stock prices accurately.

III. Solution for Problem Statement

XGBoost is a fast ML algorithm that supports out-of-core computation. It has an edge over the other ML algorithms as it can overcome memory limits, making it exceptionally fast. It can also process massive datasets to predict stock prices. It has a higher speed in the training process and improves scalability with the help of its parallel and distributed computing capabilities. The regularised objective function of XGBoost prevents overfitting while providing high predictive power, allowing predictions that can be used to make reliable stock market forecasts. Its sparsity-aware adaptive filtering algorithms can also handle missing or sparse data, a common problem in real-world applications. With a customizable objective function, the proposed algorithm would be suited for the volatile behaviour of the Indian stock market and fill the gap till a region-specific predictive task. These features make XGBoost a powerful and flexible solution, beating traditional regression algorithms and other machine-learning models.

IV. Related work

Sureshkumar et al. present the basics of stock market analysis and use various regression analysis techniques and other estimation processes to predict future stock prices. This paper also summarises the existing error and accuracy coefficients. It focuses mainly on the study of one stock, i.e., Infosys technologies. It demonstrates that isotonic regression is one of the most accurate methods for stock price prediction[1].Yang et al. propose a stock price prediction method based on XGBoost and LightGBM models that preprocesses the original data with feature engineering and conducts average processing for the missing values. Then the predicted results of three single models and a combined model were compared. Among them, the utility value of the Proposed Model was the highest (7852), the utility value of XGBoost was (6008), the utility value of LightGBM was (7487,) and the utility value of Neural Network was (5019). It is found that the XGBoost and LightGBM model with 1:1 fusion has better prediction effects than the single model and the neural network model.[2]

Özkurt et al. improve the efficiency of personal loan eligibility decisions by optimising machine learning techniques specifically, using Gradient Boosting, XGBoost, and AdaBoost algorithms. The methodology is based on the analysis of the “Bank Personal Loan Dataset”, available on Kaggle, which includes variables such as age, experience, income, family size, credit card debt shares, and whether or not defaulting on loan has been observed. These data were used to train and test 3 ML algorithms. These ML models yielded high efficiency in predicting loan eligibility, thus making approving a loan application easy without compromising the financial institution. This paper establishes gradient boosting as a very efficient algorithm due to its capability to learn complex patterns, flexibility, and robustness against overfitting. This paper deduces that XGBoost has its strengths in speed, scaling, and predicting missing values. AdaBoost achieves very good efficiency by integrating different weak learners with one reliable classifier. The downside is that AdaBoost is known to be unstable in the presence of noise and outlier observations (i.e. it does not have a sparsity-aware adaptive filter). The paper ends with a comprehension stating that with the use of Gradient Boosting, XGBoost, and AdaBoost, significant default reduction can be achieved for effective decision-making in an institution with acceptable tolerable levels. It was observed that XGBoost is the best as it gives an upper hand over the other two in terms of accuracy and precision.[3]

Sangeetha et al. propose to use an Evaluated Linear Regression (ELR) based Machine Learning (ML) approach for stock market prediction. The ML model performs linear regression on open, close, low, high and volume features to predict continuous values like the mean price and trade volume. It is concluded from this paper that the ELR-ML model provides more accurate prediction than traditional regression models[4].

Meher et al. explore one of the most common machine learning algorithms, i.e., random forest regression. In its use case in the fintech industries, it uses linear regression as its base to predict the stock market price with the help of ensemble forest algorithms with an accuracy rating of 50 % according to their research which is based on the HAR framework that they prove using high-frequency data and formulated models. The paper also argues that several optimizations make XGBoost unique. It can parallelize tree building, which speeds up training, and it is also sparsity-aware, which helps us deal with missing or sparse data. Also, for datasets larger than the available memory, out-of-core computation is used with strategies like block compression and block sharding for disk-based data without severe performance loss[5].

Chen et al. delve into the XGBoost algorithm, a gradient-boosting algorithm that uses an end-to-end scalable tree. The paper first explores the world of gradient boosting algorithms and split finding algorithms with a heavy emphasis on cache management and parallel processing to reduce the time and space complexity that can help us to analyse big data more effectively[6]. Kannan et al. explore data mining techniques to predictwhich direction (upward vs. downward) the price of stocks will move. The method combines 5 technical analysis techniques: Typical Price (TP), Chaikin Money flow

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indicator (CMI), Stochastic Momentum Index (SMI), Relative Strength Index (RSI) and Bollinger bands (BB). The performance of the proposed algorithm: BSRCTB (Combinatorial Technique) is demonstrated by a profitable signal rate of 58.25%, above the reference Moving Average (MAV) and the analysed stock return of 52.62%[7].

Wuthrich et al. use the fundamental analysis technique of extracting textual web data from various news websites and predicting stock prices by assigning weights to the different word tuples in news articles. It predicts the index levels of some Asian and some Western markets and provides the accuracy and error percentages[8].

Lu et al. explore various neural networking algorithms for stock price prediction. Amongst these algorithms, the paper concludes the CNN-LSTM (Convolutional Neural Networks - Long Short-Term Memory) is the most accurate neural networking model for predicting stock prices among the techniques considered therein[9].

Nayak et al. propose a stock price forecasting model for Indian stock markets based on past data and sentiment analysis. The framework integrates stock information like historical prices and trading volumes with integrated sentiments procured through the analysis of headlines of news articles and social media. Sentiment analysis is an additional feature for predicting stock price movements daily and monthly. One of the model's weaknesses is that it depends on accurate and timely sentiment data. In other words, the accuracy of the level of precision of the predictions is determined by sentiment analysis. The authors conclude the paper by showcasing the applicability of their model in forecasting the movements of stock prices in the Indian stock market. The model records more than 70% accuracy on monthly forecast prediction[10].

Shashidhar et al. explore the possibility of using news sentiment analysis paired with Fast Fourier Transform (FFT) and Inverse Fast Fourier Transform (IFFT). This study uses the Yahoo Finance (yfinance) API data to predict stock prices using regression models. It experiments with regression models like Linear Regression, Lasso Regression, Ridge Regression, and Support Vector Regression. The paper concludes that public sentiment analysis is useful for stock price prediction and should be used with previous stock price data to accurately predict future stock prices[11].

Kumar et al. evaluate the various methods available for predicting and representing stock market movements. The methodology includes analysis of seven different statistical and pattern recognition techniques such as ARIMA, ANN, LTM, and KNN. Visualizations and forecasts help traders make informed decisions fairly quickly and help easily organise a large and complex body of data for proper information retrieval. The paper concludes that data representation simplifies the decision-making process of the traders, while systematically sound models lead to better stock selection[12].

V. Architectutre

The model architecture of the proposed system can be divided into sections/processes as under:

A. Data Collection

Data collection is the systematic approach to collecting, measuring, and analysing information from various sources to gain insights, make informed decisions or train models for use in research, business or technology.

B. Data Preprocessing

Data Preprocessing is defined as preparing the data for use in machine learning algorithms. Here it is verified that the data grows consistently and error-freely to be used by the algorithm. Depending on the data and task requirements, Data Preprocessing usually involves 2 subtypes:

1) Data Cleaning

Data cleaning eliminates incorrect, incomplete, blank or duplicate data from a dataset.

2) Data Slicing

Data slicing is the process wherein data is sliced or separated into parts for training and testing an ML model.

C. Dash and Flask Framework

Dash is a Python web framework that enables users to create interactive and data-heavy web applications almost seamlessly. Dash combines Flask, React and Plotly.js to ease the building of highly interactive and data-driven applications. In Dash applications, the two elements that are the focus of design are layout and interactivity. The layout is a Python representation of the HTML elements and their attributes. The components, which include graphs, sliding, dropdowns, and buttons, are displayed on the browser through Plotly's sophisticated graphic visualisations.

Flask web server is a micro web framework web developing tool that provides a headroom for advanced applications. Flask is a WSGI-compliant web server, i.e., it can work flawlessly with other web servers.

VI. Methodology

The data for this study was sourced from the yfinance (Yahoo Finance) API from 1995 up to and including November 30, 2024. After that, the following procedure is used to train the data. Multi-parameter input forms one of the most important bases of XGBoost. In the base/weak learners' set, the parameter should have been hyper-tuned to avoid cases where there is overfitting.[17] According to these input parameters, the model produces a function. The gradient of the function is estimated, and then the model is set to its default value after which the gradient is calculated. The change of function thus created is known as the gradient function and the second derivative is known as Hessian. This model undergoes further tuning after fitting gradient and hessian functions onto weak or basic learners over a training set.[6]

Mathematically, the XGBoost algorithm is represented as:

$$\hat{y_i} = \phi(x_i) = \sum_{k=1}^{K} f_k(x_i),\ f_k \in F$$

where,

- K is the number of trees,
- f is the functional space of F, and
- F is the set of possible classification and regression trees.

The objective function for the model is given by:

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$$L(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k)$$

where the first term is the loss function, the second term is the regularisation parameter and

$$\Omega(f) = \gamma^T + \frac{1}{2}\lambda||w||^2$$

The loss function accounts for the inaccuracy in the model by introducing an error correction into the model to minimise prediction errors. The regularisation parameter controls the complexity of the model to avoid overfitting.[14] Instead of learning the tree all at once, the loss that it has learned and adds a new tree at each iteration. Hence, the objective function is:

$$L^{(t)} = \sum_{i=1}^{n} l\left(y_i, \widehat{y_i^{(t-1)}} + f_t(x_i)\right) + \Omega(f_t)$$

The predicted data is sent to plotly as input parameters to display the graph.

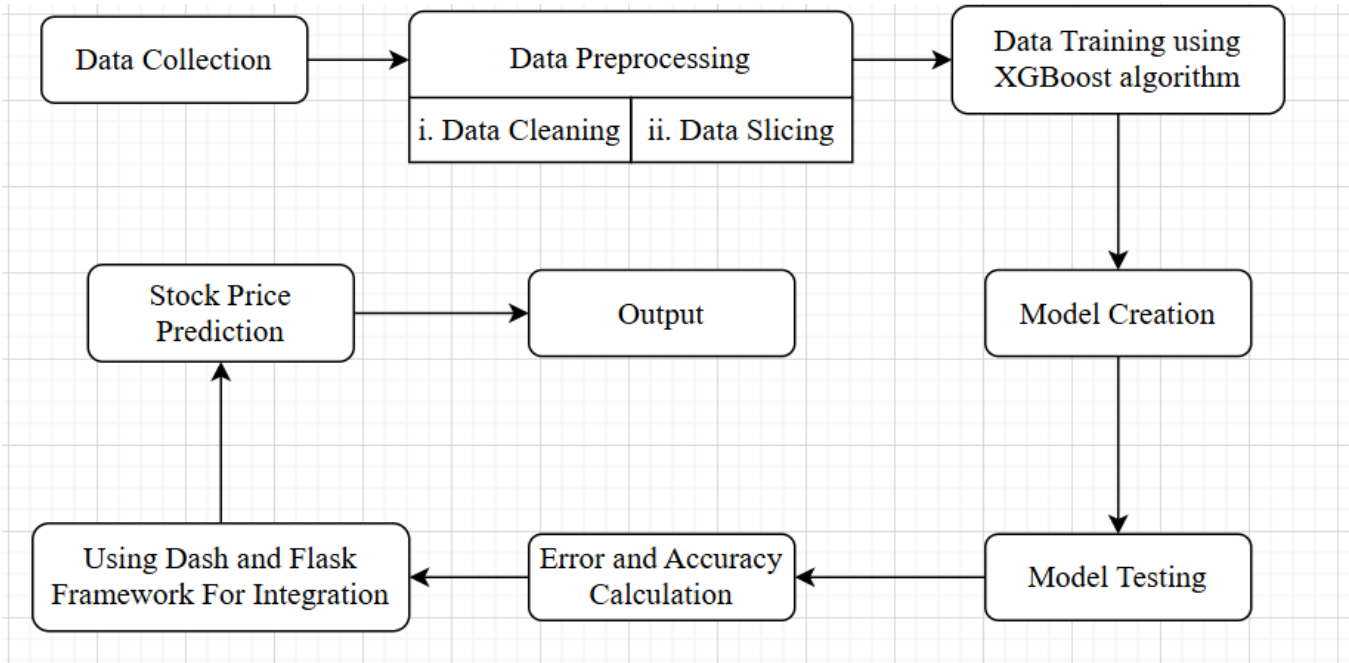

Fig. 1. Proposed System Architecture

VII. Result and Analysis

Table 1 tabulates the predicted values of our model and the actual values of four stocks, Infosys, SBI, ICICI, and TCS, from 3 December 2024 to 6 December 2024.

The Root Mean Square Error (RMSE) was calculated using the following formula:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(x_i - y_i)^2}{n}}$$

The Mean Absolute Error (MAE) was calculated using the following formula: $MAE = \frac{1}{n}\sum_{i=1}^{n} |x_i - y_i|$

The results of these error calculations have been tabulated in Table 2. Fig. 2 shows the line chart of the prediction price and the actual price of TCS stock as per the data in Table 1.

The formula for Accuracy is:

$$Accuracy\ Percentage = 100 - ERROR\ Percentage$$

Table 3 shows the corresponding accuracy calculations.

TABLE I. Predicted and Actual Prices of Stocks

	Infosys		SBI		ICICI		TCS	
Date	*Predicted*	*Actual Price*	*Predicted Price*	*Actual Price*	*Predicted Price*	*Actual Price*	*Predicted Price*	*Actual Price*
03-Dec-24	1899.14	1890	825.23	853.5	1300.99	1306.75	4276.66	4303
04-Dec-24	1900.56	1892	850.36	858.55	1308.55	1315.6	4302.66	4343.1
05-Dec-24	1922.45	1935	862.65	864.25	1313.05	1336.55	4465.05	4455.15
06-Dec-24	1946.09	1919	865.04	862.95	1320.88	1329	4440.44	4444.15

TABLE II. Error Calculation

	Infosys	SBI	ICICI	TCS
RMSE(%)	16.188	14.775	13.239	24.773
MAE(%)	14.335	10.0375	11.1075	20.1625

TABLE III. Accuracy Calculation

	Infosys	SBI	ICICI	TCS
RMSE(%)	83.812	85.225	86.761	75.227
MAE(%)	85.665	89.9625	88.8925	79.8375

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TABLE IV. ACCURACY COMPARISON

	Linear Regression	Pace Regression	Simple Linear Regression	Senti-mental Analysis	XGBoost(Pro-posed System)
Accuracy	61.167	61.189	61.167	70	84.42

Table 4 compares the accuracy of the proposed system to the accuracies of the different models found in the literature survey.

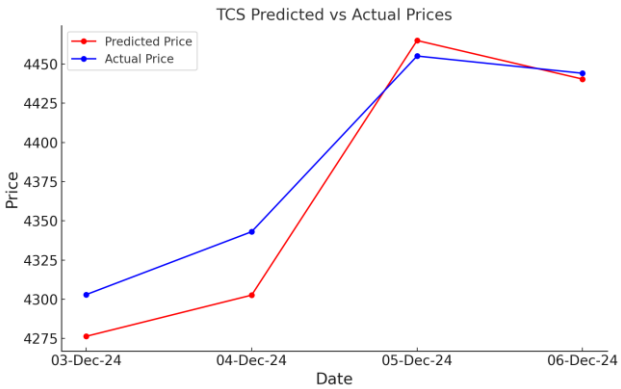

Fig. 2. TCS stock price prediction

The average error percentage using this model by RMSE method comes out to be 17.244% and hence, the accuracy percentage as calculated by the above-given formula is 82.756% while the average error percentage using this model by MAE percentage comes out to be 13.911% and hence, the accuracy percentage turns out to be 86.089%.

VIII. CONCLUSION

XGBoost as a predicting algorithm increases its accuracy over several iterations where it is found that it can surpass the accuracy of even some DNN models.[15] It is also one of the most consistent boosting models.[16] Extreme Gradient oosting (XGBoost) applies the concept of gradient boosting but improves its speed, scalability and precision[17] XGBoost is one of the best machine learning algorithms as long as hybrid models are not considered. It is superior to traditional machine learning models such as logistical regression, bagging etc.[18] Advanced regularized objective functions plus second-order derivatives (Hessian) significantly minimize errors while reducing overfitting. It can parallelise tree building, which speeds up training, and it is also sparsity-aware, i.e., it deals with missing or sparse data, out-of-core computation, and block compression. This study concludes to have an accuracy of approximately 84.42% on average, using only the XGBoost algorithm. Hence, it is one of the best ML techniques.

IX. FUTURE SCOPE

Work in this area, in future, may entail some neural networking techniques in conjunction with the XGBoost algorithm to make the stock price prediction even more accurate. XGBoost when incorporated with LightGBM can enhance the predicting output which can give highly accurate predictions.[2] XGBoost is an adequate prediction algorithm but its accuracy can be further enhanced using Neural Networks such as CNN, KNN and ANN.[19,21,22]

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